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Ex 15: Decision Tree

CODE:

```
# Experiment 15: Decision Tree (Play Tennis Predictor)


print("--- Enter Weather Condition ---")
print("1: Sunny")
print("2: Overcast")
print("3: Rain")
weather = int(input("Enter Weather choice (1/2/3): "))


if weather == 2:
  # Overcast branch: always Yes
  decision = "Yes (Play Tennis)"
elif weather == 1:
  # Sunny branch: depends on Humidity
  print("\n--- Enter Humidity Level ---")
  print("1: Normal")
  print("2: High")
  humidity = int(input("Enter Humidity choice (1/2): "))
  if humidity == 1:
    decision = "Yes (Play Tennis)"
  else:
    decision = "No (Do not play)"
elif weather == 3:
```

```
    # Rain branch: depends on Wind
    print("\n--- Enter Wind Condition ---")
    print("1: Weak")
    print("2: Strong")
    wind = int(input("Enter Wind choice (1/2): "))
    if wind == 1:
        decision = "Yes (Play Tennis)"
    else:
        decision = "No (Do not play)"
else:
    decision = "Invalid Input"

print("\nResult:", decision)
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-15.py ===========
--- Enter Weather Condition ---
1: Sunny
2: Overcast
3: Rain
Enter Weather choice (1/2/3): 1

--- Enter Humidity Level ---
1: Normal
2: High
Enter Humidity choice (1/2): 2

Result: No (Do not play)
```